Variation of parameters for nonhomogeneous equations

Like the method of undetermined coefficients, variation of parameters is a method we can use to find the general solution to a second order nonhomogeneous equation.

Variation of parameters works for a wider family of $g(x)$ functions than undetermined coefficients, which makes it more versatile. But variation of parameters also requires some integration, and there's no guarantee that we'll get functions that are easy to integrate.

Variation of parameters

Just like with undetermined coefficients, we'll start by substituting $g(x) = 0$ to change the nonhomogeneous equation into a homogeneous equation. Then we'll use the associated characteristic equation to find the complementary solution $y_c(x)$.

Handling the particular solution is where the method of variation of parameters starts to diverge from the method of undetermined coefficients.

To find a particular solution using variation of parameters, we'll start by picking out the fundamental set of solutions $\{y_1, y_2\}$ from the complementary solution. This set of solutions will be everything but the coefficients c_1 and c_2.

Complementary solution $y_c(x)$ **Fundamental solution set** $\{y_1, y_2\}$

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$$y_c(x) = c_1 e^{r_1 x} + c_2 e^{r_2 x} \qquad \{y_1, y_2\} = \{e^{r_1 x}, e^{r_2 x}\}$$

$$y_c(x) = c_1 e^{r_1 x} + c_2 x e^{r_1 x} \qquad \{y_1, y_2\} = \{e^{r_1 x}, x e^{r_1 x}\}$$

$$y(x) = e^{\alpha x}(c_1 \cos(\beta x) + c_2 \sin(\beta x)) \qquad \{y_1, y_2\} = \{e^{\alpha x} \cos(\beta x), e^{\alpha x} \sin(\beta x)\}$$

Once we've identified the fundamental set of solutions, we'll plug y_1 and y_2 and their derivatives into this system of linear equations:

$$u_1' y_1 + u_2' y_2 = 0$$

$$u_1' y_1' + u_2' y_2' = g(x)$$

We'll solve the system for u_1' and u_2', then integrate to find u_1 and u_2. This is the integration we mentioned earlier. Depending on what we find for u_1' and u_2', it won't always be easy, or even possible, to integrate to u_1 and u_2.

But if we're able to get u_1 and u_2, then we can say that a particular solution is

$$y_p(x) = u_1 y_1 + u_2 y_2$$

And just like with undetermined coefficients, the general solution to the second order nonhomogeneous equation is still the sum of the complementary and particular solutions,

$$y(x) = y_c(x) + y_p(x)$$

One final note. With this method, it's important that we start with a coefficient of 1 on y''. If our differential equation has a y'' term with a coefficient that's anything other than 1, then our first step should be to

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divide through the equation by that value to change the coefficient on y'' to 1.

Let's do an example so that we can see exactly how we can use variation of parameters to get to the general solution.

Example

Use variation of parameters to find the general solution to the differential equation.

$$y'' + 4y' + 4y = \frac{e^{-2x}}{x^3}$$

The coefficient on y'' is already 1. We'll substitute $g(x) = 0$ to rewrite the nonhomogeneous equation as a homogeneous equation.

$$y'' + 4y' + 4y = 0$$

Now we'll use the associated characteristic equation to find roots.

$$r^2 + 4r + 4 = 0$$

$$(r + 2)(r + 2) = 0$$

$$r = -2$$

We get equal real roots, so the complementary solution is

$$y_c(x) = c_1 e^{-2x} + c_2 x e^{-2x}$$

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and the fundamental set of solutions is therefore

$$\{y_1, y_2\} = \{e^{-2x}, xe^{-2x}\}$$

If we differentiate the set, we get

$$\{y_1', y_2'\} = \{-2e^{-2x}, e^{-2x} - 2xe^{-2x}\}$$

Create a system of linear equations using the fundamental set and its derivatives.

$$u_1'y_1 + u_2'y_2 = 0 \qquad u_1'y_1' + u_2'y_2' = g(x)$$

$$u_1'e^{-2x} + u_2'xe^{-2x} = 0 \qquad u_1'(-2e^{-2x}) + u_2'(e^{-2x} - 2xe^{-2x}) = \frac{e^{-2x}}{x^3}$$

$$-2u_1'e^{-2x} + u_2'e^{-2x} - 2u_2'xe^{-2x} = \frac{e^{-2x}}{x^3}$$

Our next step is to solve this system for u_1' and u_2'. We'll start by multiplying the first equation by 2,

$$2u_1'e^{-2x} + 2u_2'xe^{-2x} = 0$$

$$-2u_1'e^{-2x} + u_2'e^{-2x} - 2u_2'xe^{-2x} = \frac{e^{-2x}}{x^3}$$

then we'll add the equations to eliminate u_1'.

$$(2u_1'e^{-2x} - 2u_1'e^{-2x}) + (2u_2'xe^{-2x} - 2u_2'xe^{-2x}) + u_2'e^{-2x} = 0 + \frac{e^{-2x}}{x^3}$$

$$u_2'e^{-2x} = \frac{e^{-2x}}{x^3}$$

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$$u_2' = \frac{1}{x^3}$$

Use this value for u_2' to find u_1'

$$u_1' e^{-2x} + u_2' x e^{-2x} = 0$$

$$u_1' e^{-2x} + \left(\frac{1}{x^3}\right) x e^{-2x} = 0$$

$$u_1' e^{-2x} + \frac{e^{-2x}}{x^2} = 0$$

$$u_1' e^{-2x} = -\frac{e^{-2x}}{x^2}$$

$$u_1' = -\frac{1}{x^2}$$

Integrate u_1' to find u_1,

$$u_1 = \int u_1' = \int -\frac{1}{x^2}\, dx$$

$$u_1 = \int -x^{-2}\, dx$$

$$u_1 = -\frac{1}{-1} x^{-1}$$

$$u_1 = x^{-1}$$

$$u_1 = \frac{1}{x}$$

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and then integrate u_2' to find u_2.

$$u_2 = \int u_2' = \int \frac{1}{x^3}\, dx$$

$$u_2 = \int x^{-3}\, dx$$

$$u_2 = \frac{1}{-2}x^{-2}$$

$$u_2 = -\frac{1}{2}x^{-2}$$

$$u_2 = -\frac{1}{2x^2}$$

Given these values for u_1 and u_2, the particular solution is

$$y_p(x) = u_1 y_1 + u_2 y_2$$

$$y_p(x) = \frac{1}{x}e^{-2x} - \frac{1}{2x^2}xe^{-2x}$$

$$y_p(x) = \frac{e^{-2x}}{x} - \frac{e^{-2x}}{2x}$$

$$y_p(x) = \frac{2e^{-2x} - e^{-2x}}{2x}$$

$$y_p(x) = \frac{e^{-2x}}{2x}$$

Adding this particular solution to the complementary solution gives us the general solution $y(x)$.

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$$y(x) = c_1 e^{-2x} + c_2 x e^{-2x} + \frac{e^{-2x}}{2x}$$

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